

Data Analyst

Technical Challenge

Welcome to the Data Analyst challenge

The objective of this task is to assess your thinking process when doing data analysis. We want to understand how you approach data. We know there are multiple ways to solve a problem, therefore we are not expecting a *single right answer*.

What matters most is your thought process and how you present your results.

What we expect

Here are some guidelines to keep in mind while creating your output:

- Professionally written, succinct, clear with actionable insights. **We want to understand and how you present and communicate your findings.** Imagine the CEO will read it.
- Send the insights, all code, and any other supporting files inside a zipped folder back to us (.zip format). **We want to understand how you arrived at your insights.**
 - Please do not submit files in any particular BI tool format (e.g. .PBIX,.TWB, etc...). Assume we won't be able to open it.
 - Python/R files can be submitted in any format.
- **You only have 7 days to put a presentation together, we don't expect you to explore all aspects of the data just enough to present for 20 minutes.**

Task Introduction

The company has just launched a new 2 player puzzle board game that's now available in the Play and App Store.

This is an online mobile game where users can play against other players or against a bot, depending on the game mode selected:

- classic - play against another player, giving a total of 10 minutes per player to make their moves. If a player timer runs out they automatically lose regardless how far in the board they were.
- quick - play against another player, giving a total of 5 minutes per player to make their moves. If a player timer runs out they automatically lose regardless how far in the board they were.
- bot - play against the computer, timing out after 10 minutes if no one had won until then

In order to win a match the users have to be the first ones reaching the final mark on the board.

On the game there are also several products available that the users can buy with real-world money. This products can vary from:

- Gems and Coins Packs, needed to purchase other products in game
- varied Bundles, that give the users different products such as customized pawns and boards
- Season Pass, a feature that has a road full bunch of prices (from gems, coins, bundles) where users need to play matches to progress in the road and collect the prices

In order to understand its performance before making it globally available, this new game is currently only available in Brazil, India and United Kingdom and United States. Therefore, **what insights and recommendations will you give the CEO about this new game?**

Data

You can find the data sets for this task in a zipped folder **challenge_task_data.zip**. Inside the zip file, you will find three data sets:

1. **data_daily_activity.csv** containing the daily activity of the users. Each row represents a user that joined the game on that specific day. This data set has the following structure:
 - *game* - The name of the new game
 - *activity_date* - The calendar date when the user was active
 - *acquisition_date* - The calendar date when the user was active in the game for the first time
 - *platform* - The users' reported platform on each calendar date
 - *country* - The users' reported country on each calendar date
 - *acquisition_type* - The way the user installed the game
 - *user_id* - The unique user ID of each active user
2. **data_matches.csv** containing the all the matches played during the period of the analysis per user. Each row represents match. This data set has the following structure:
 - *game* - The name of the new game
 - *activity_date* - The calendar date when the user was active
 - *user_id* - The unique user ID of each active user
 - *game_mode* - The game mode where the user played
 - *end_reason* - The result of the matches
 - *n_matches* - The total number of matches per user by calendar date, game mode and end reason
3. **data_in_app_purchases.csv** containing the all the in-app purchases made during the period of analysis per user. Each row represents match. This data set has the following structure:
 - *game* - The name of the new game
 - *activity_date* - The calendar date when the user was active
 - *user_id* - The unique user ID of each active user
 - *purchase_number* - The number of the purchase made by the specific user_id
 - *product_group* - The group of the product purchased
 - *product* - The name of the product purchased
 - *dollar_purchase_value* - the revenue that purchased generated in dollars

Additional Info

- Due to the high volume of applications we do not provide feedback on individual submissions.
- If you make it through to the next round you will be able to present your finding.
- Due to the high volume of submissions please be patient about hearing back from us. We will try our best to get back to you as soon as possible.